

Small Models by Pruning and Distillation

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Abstract

Models are usually trained from scratch at every size. We prune a large model and retrain the smaller ones with a fraction of the data, and they match or beat models trained from scratch at the same size, at a fortieth of the cost.

1 Introduction

Large models dominate language processing [1, 2]. Training a family of them from scratch is costly, so we prune the largest and retrain the rest with distillation [2]. This paragraph goes on long enough to fill several lines of running text, so that the body of the page is measured properly by the reader, and it cites more work [1] as it goes.

1.1 Importance

Estimating the importance of neurons, heads and layers is well studied [1, 2]. Recent work on pruning language models has focused on gradients [1], cosine similarity [2] and perplexity [1] on a calibration set, which we compare here. We define the layer as $\text{MLP}(X) = \delta(XW_1^T)W_2$, where $W_1, W_2 \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{model}}}$, and d_{model} and d_{hidden} are the embedding and hidden dimensions of the model we prune in the rest of this paper.

2 Results

We compare the pruned models with models of their size in the two tables below, and give the chat results after them.

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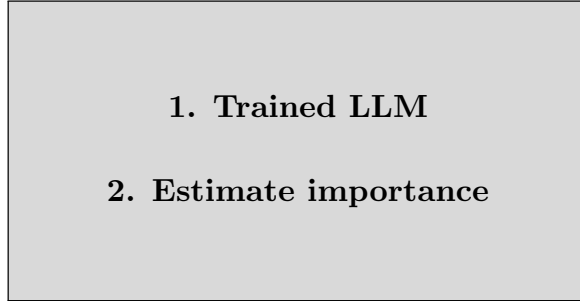


Figure 1: Overview of the approach: the trained model, and the importance estimated.

	Benchmark	Metric	Models		
			Base	Mistral	Ours
Code Logic	# Parameters		8B	7.3B	8.3B
	# Tokens		15T	8T	94B
	winogrande (5)	acc	77.6	78.5	79.0
	MMLU (5)	acc	65.3	64.1	63.8
	MBPP (0)	pass@1	42.4	38.8	35.2
	HumanEval (0)	pass@1	28.1	28.7	31.6

Table 1: Performance of the pruned 8B model against models of its size.

A Appendix

A.1 One-shot vs. Iterative Pruning

Loss components	LM loss
logits	2.155
logits + embedding	2.141

Table 6: Ablation of the loss components.

References

- [1] A. Author, “A first paper,” 2023.
- [2] B. Writer, “A second paper,” 2024.

Benchmark	Phi	Gemma	Ours
MMLU (5)	57.5	42.0	58.6
HellaSwag (10)	75.2	72.0	75.0

Table 2: Performance of the pruned 4B model against models of its size.

Model	Non-Emb. Params	Total
Ours-instruct	2.6B	6.46
Phi-chat	2.5B	4.29

Table 3: Results on MT-Bench.

Model	Avg.
Ours-instruct	53.09
Gemma-it	41.63
Llama-instruct	50.51

Table 4: Results on BFCL.

DEP	MLP	ATT	EMB	Distillation Loss	LM Val Loss
✓	✓	✓	✓	5.35 $\rightarrow$ 0.38	2.062
×	✓	✓	×	5.07 $\rightarrow$ 0.42	2.101
Train from scratch (random init)				12.27 $\rightarrow$ 2.34	3.953

Table 5: Pruning strategies before and after retraining.